Class I
Transportation Engg

1) Highway

2) Railway

3) Dock & harbour

4) Airport & tunnel

Highway Engg →

Highway engg—

1) Introduction

2) Geometric design.

3) Sight distances

4) traffic engineering

5) Material engineering & material survey

6) Road characteristic.

7) Intersection or signal design.

8) Highway capacity & Miscallaneous.

Highway Engg

Introduction

Highway developement & plan

Roads kis order me develop hui?

1) Roman Roads
2) Tresguard
3) Metcalfe
4) Telford road method
5) Macadam road method
6) CBR method.

(sequence is imp)

Road developement committees in India.

kya kya committe hai road ke developement ke liye india me

1) Jaykar committee. – 1927
2) IRF (Central road fund) – 1929
3) Indian road congress IRC (1934)
4) Motor vehicle act. MVA (1939)
5) 1st 20 year road plan.
6) CRRI [Central road research institute] 1950
7) NHA National highway act
8) 2nd 20 year road plan
9) Highway research board.

20 yr road plan

- 1st
  - year
  - Name
  - target
  - Names of road given
  - Pattern?
- 2nd
  - Year
  - Name
  - kya Introduce kiye?
  - target
- 3rd
  - Year
  - Name
  - Roads classification kaise kiya
  - target

20 year road plan

1st 20 yr road plan.

- 1943 - 1963
- Nagpur road plan.
- length 16 km / 100 km² area.
- NH, SH, MDR, ODR, VR
  - NH, SH: major
  - MDR, ODR: district road
  - SH: state highway
  - ODR: other
  - VR: village road
- Star & grid pattern / Radial & grid pattern

2nd 20 yr road plan

- 1961 - 1981
- Bombay road plane
- Expressway introduced + NH SH MDR OR VR
- 32 km / 100 km² area.

3rd 20 yr road plan.

- 1981 - 2001
- Lucknow road plan
- primary 1° < EW, NW
- secondary 2° < SH, MDR
- tertiary 3° ODR, VR

Requirement of ideal road kya hai.

Highway align karne ke liye kya kya survey karte hai.

Kis type ke survey hai.

Best alignment kaunse survey me final karte hai.

Isme 3m soil survey kaunse me karte hai.

Kis survey me ... poora soil profile banate hai.

DPR ke kya kya components hai.

Requirements of ideal roads

Short safe easy & economical.

Surveys for highway alignment

1. Map study
2. Reconaissance
3. Preliminary survey
4. Detailed survey & final location survey

Engg survey

Finalisation of best alignment is done in preliminary survey.

② Soil survey from 1m to 3m is done in preliminary survey

③ In detailed soil survey complete profile of soil is prepared.

Components of DPR. (Detailed proj report)

① Report. ② estimate. ③ Drawings

1.6

Roadway
- other name
- formula.
- Right of way
  - name
  - formula.

Roadway or formation width.

Roadway = carriageway + shoulder

Road land or right of way or landwidth = Roadway + road margin.

Road with in m
- Single lane road
- two lane road
- without raised kerb
- with raised kerb
  - generally kaha detehi
- multilane pavement
- Intermediate carriageway
- width of vehicle.
- Single deck vehicle ht
- double decker vehicle ht.

Road width / pavement width (in general)

1. Single lane road : 3.75 m
2. two lane road
   without raised kerb = 7m
   with raised kerb = 7.5m.
   (generally used on bridges)
3. Multilane pavement : 3.5 m per lane
4. Intermediate carriageway : 5.5 m

Width of vehicle in general = 2.44m

Single deck vehicle ht = 3.8m.

Double deck vehicle ht = 4.7m

Roadwidths for	plain & rolling terrain	hilly & mountaneous / steep terrain.
1. NH-single & SH lane	12 m	6.25 m
. double lane	12m	8.80 m.
2. MDR . single lane	7m	4.75 m
. double lane	9m	—
3 ODR . single lane	7.5	4.75m
. double lane	9.0	—

(1.9) Crossectional elements of road

- Caml/cs
- Sh
- Med./div
- Ke
- Wi

Crossectional elements of pavement or road

1) camber / cross slope
2) Shoulder
3) Median / divider.
4) Kerb.
5) Width of pavement

Cross slope/Camber

- purpose.
- Gradient camber relation?
- Camber values for heavy rfall & light rfall

Camber or cross slope.

- it is provided to drain off rain water or remove water to prevent entry of surface water into pavement layer.
- G = 2C. gradient = twice of camber

	Heavy rainfall	light rainfall
① CC road & high bituminous road	1:50	1:60
② Thin bituminous road	1:40	1:50
③ WBM & Gravel road	1:33	1:40
④ Earth road	1:25	1:33

Surface roughness Index
- Bump integ
  - Good pavement ki value.
  - purpose
  - Result kaise deta h?
- International roughness index.
  - kya purpose h ye index ka?

Surface roughness index

1) Bump integrator

→ Good pavement ke liye 150 cm/km hona chahiye value

. The unevenness of pavement surface is commonly measured by using a simple equipment called bump integrator.
. Result it gives vertical undulation of pavement per unit length of road.
Unit → cm/km

2) International roughness index.

. Riding quality of pavement surface is quantified in terms of roughness index IRI.

Sight distances
- SSD
  - other name
  - what is it?

Sight distances.

SSD: Stopping sight distance or non passing sight distance.
. It is summation of lag or reaction distance + breaking distance

Class 2 4 August 24

SSD
- lag distance kya hai
- Reaction time SSD mein kya role hai
- Brake distance
  - Definition
  - kinetic energy kise equal honi chahiye.
- PIEV Theory

lag distance.
lag distance is the distance covered or travelled by vehicle during PIEV reaction time.

time taken reaction time in SSD : 2.5 seconds.

Brake distance : It is the distance travelled by vehicle after application of brake.

- kinetic energy lost should be equal to work done by vehicle.

PIEV theory

1) Perception
2) Intellection
3) Emotion
4) Volition.

SSD

- lag distance ka formula kya h?
  - $1\ ms^{-1}$ in km^2
  - lag dist formula in kmph?
- Breaking distance
  - formula in ms^{-1}
  - formula for ascending/descending gradient
  - formula in kmph for gradient

(1) Lag distance (S_l)

S_l : velocity × reaction time

$= [V t_R]$ → sec, → m/s

$1\ ms^{-1} = 0.278$ kmph

$= 0.278\ V t_R$ → sec, → kmph

(2) Braking distance (S_b)

work done = kinetic energy loss

$$(W.f) * S_b = \frac{v^2}{2g}$$

$$S_b = \frac{v^2}{2gf} \quad v \rightarrow ms^{-1}$$

$$S_b = \frac{v^2}{2g(f \pm N\%)}$$ ← gradient

$$S_l = \frac{V^2}{254[f \pm N\%]}$$ V → kmph; N → gradient

N is +ve for ascending gradient

N is -ve for descending gradient

f → longitudinal friction coefficient

2.3) SSD

- final formula

- f ki range & one val.

SSD : $S_1 + S_2$

$SSD = 0.278 V t_R + \frac{V^2}{254(f \pm N)}$

↳ brake efficiency coeff

longitudinal friction

range : 0.35 to 0.40

= 0.35

Note : for 1 lane one way road = SSD

- 1 lane 2 way road ke liye SSD kitna lenge?

for 1 lane two way road

= 2 SSD

- 2 lane 2 way ccd?

for 2 lane 2 way road

= SSD

- ISD?

Intermediate sight distance

= 2 SSD

- HLSD?

Headlight sight distance

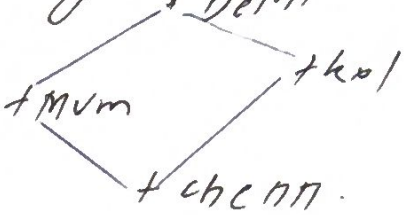
= SSD

NHDP
me kya
kya hai

Imp facet pts

NHDP : National highway
developement project / programme

1) Golden Quadrilateral



2) Corridor.

(i) N-S corridor [Srinagar - Kanyakumari]

(ii) E-W [Silchar to porbandar]

3) Port connectivity

NRRDA.
- full form.
- PMGSY
  - launch date
  - target

NRRDA : National rural road developement Agency

(i) PMGSY :
Dec 2000

target : population of more than 1000 should be connected to good all weather road. till 2003

2.5

- P7500 ke lie PMGSY me kya tha?

- 3rd 20 yr road plan

- length of NH formula

- length of SH formula

- length of MDR

- Expressway target.

. P > 500 should be connected to above population villags upto 2007.

Third 20 year road plan (Lucknow road plan)

length of NH : $\frac{\text{Area in km}^2}{50}$

length of SH = $\frac{\text{Area in km}^2}{25}$ or

62.5 x No of towns

length of MDR = $\frac{\text{Area in km}^2}{12.5}$

: 30 x No. of towns.

Expressway target : 2000 km

NHAI

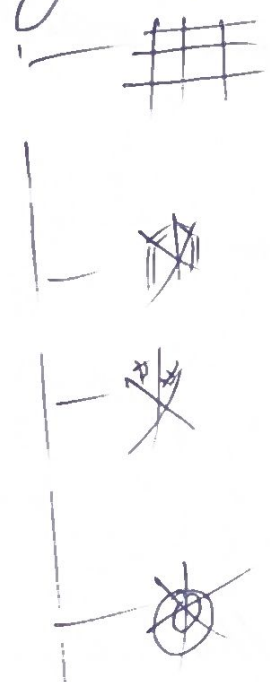
- established yr
- Operatnl yr

NHAI (National highway authority of India). 2.6

. Established 1988

Operational 1995

Road pattern kya kya hai



Road pattern.

1) Rectangular & block pattern
Example Chandigarh.

2) Star & block pattern

3) Star (Radial) & grid pattern.
Eg Nagpur road plan

4) Star (Radial) & circular pattern.
Eg Connaught place Dilli

- Hexagonal kiske liye acchha

5) Hexagonal pattern. = Good for 3 roads meeting @ built up area.

2.7

Concept of Sat
of Syc

- alternate name
- kaise optimum length calculate karte hai

Concept of saturation system

or

maximum utility system

- In this system the optimum road length is calculated on the basis of concept of obtaining maximum utility per unit length of road.

Drawings required

- keymap
  - kya hai
- Index map
  - kya batata hai

Drawings required

① keymap

The map which shows the proposed & existing roads & imp places to be connected.

② Index map

shows the general topography of the area.

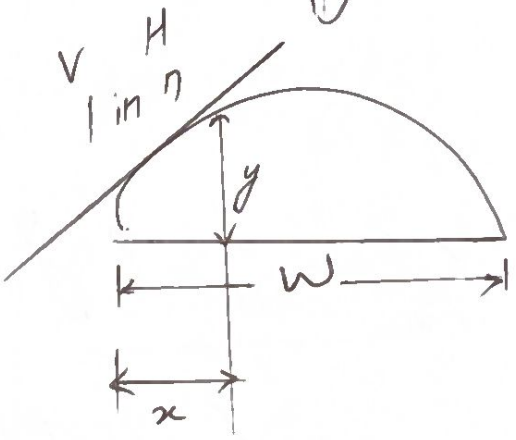
Camber
- purpose
- Shapes
  - Eliptical
    - Good for
    - formula
  - formula
- y kya h?
- W kya h?

Camber or cross slope

- It prevents entry of surface water or it is provided to drain off rain water

- Shape or types of camber
1) Elliptical or parabolic shape.
Good for bitumin roads or for roads where fast moving vehicles travel

$$y = \frac{2x^2}{nW}$$



V	H
1	n
W/2	nW/2

$y \propto x^2$

$$y = \frac{x^2}{nW/2} = \frac{2x^2}{nW}$$

y is rise or height of camber @ x horizontal.

W is width of road.

$$\text{Rise} = \frac{2 \times \text{Horizontal}^2}{\text{Slope} \cdot \text{Width}}$$

2.7 Camber

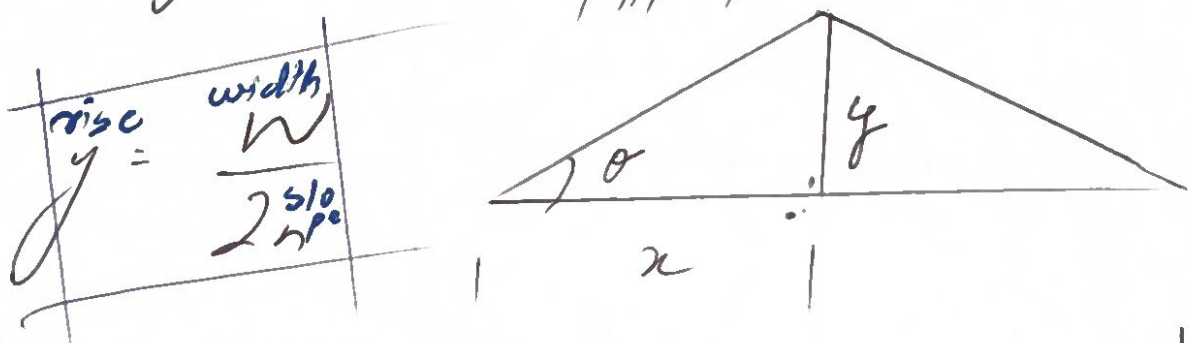
- elliptical
  - Disadvantage
    - In this shape vehicles prefer to occupy central portion of roadway hence reduces traffic capacity
  - advantge
    - comfortable
- Stright line camber

linear camber or straight line camber

- Generally used for?

generally adopted for c/c road

- formula

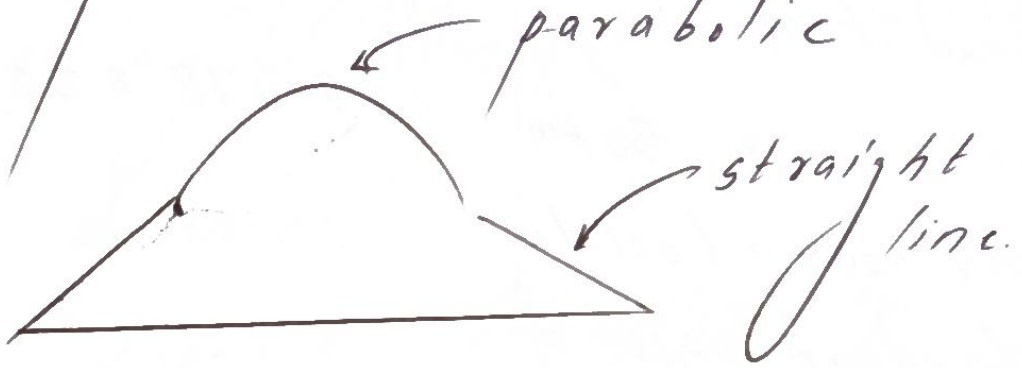


$$\tan\theta = y/x = \frac{1}{n} \Rightarrow y = \frac{x}{n} = \frac{W}{2n}$$

- advantage
  - gives good comfort for slow moving vehicles.
- Disadvantage
  - Discomfort for fast moving vehicle.

Camber shapes
- combinn

Combination of straight line & parabolic camber.



Traffic seperator
- alternate names
- Minimum desirable width.
- Min width bridges.

Traffic separator or median.

- divider / median.
- Minimum desirable width: 5 m.
- for bridges 1.2 m.

Acceleration
- formula for accelern
- formula for retardation

Acceleration or Retardation. / Deceleration.

acceler^n $a = fg$

deceler^n $a = -fg$

2. 2·11

Calculate SSD for 1 lane 2 way road if speed of vehicle is 50 km per hour. Consider standard value of coeff of friction & drivers reaction time.

Ans: $SSD = 0.278 V t_R + \frac{V^2}{254(f \pm N/100)}$

$= 0.278 \times 50 \times 2.5 + \frac{50^2}{254 \times 0.35}$

$= 62.87$ m.

1 lane 2way ke liye

$Ans^{SSD} = 2 \times 62.87 = 125.6$.

Tyre Skid

- kyo hota h?
- formula.

Tyre skid

- Due to sudden application of brake skid occurs
- formula $Sl(\%) = \frac{V_v - V_T}{V_v} \times 100\%$.

Tyre slip

- Defn.
- formula

V_v is vehicle velocity (ms^{-1})

V_T is circumferential or tyre velocity

Tyre slip.

- Application of sudden acceleration.
- formula: $S_s(\%) = \frac{V_T - V_v}{V_T} \times 100\%$.

OSD

- alternate names
- di ki value kyah m s⁻¹ me
- t ki value kyah?

OSD (passing sight distance/ non stopping sight distance)

overtaking sight distance is summation of three distances

① d_1 : distance travelled by overtaking vehicle (A) while travelling with a speed of overtaken vehicle (B). = $V_b t$: $0.278\, V_b t$ in kmph.
V_b in $m s^{-1}$

t is reaction time in OSD = 2 sec.

② d_3 is distance travelled by opposite coming vehicle in overtaking operation time.

$d_3 = V_c T$ or $0.278\, V_c T$ in kmph.

- d_3
  - kyah?
  - formula kyah?

③ d_2 is distance travelled by overtaking vehicle after accelerating & coming into the front position from B.

$d_2 = 4t(t-s)$

- d_2
  - kiska dist h
  - formula

2.13) OSD

- d_2
  - s formula
    - Default value

s is spacing b/w vehicles (overtaking & overtaken)

$s = (0.7 V_b + l)$

l is length of wheel base.

$s = 6.1\ m$.

- b
  - kya h?

b : distance travelled during acceleration.

- final formula kya h?

$d_2 = 2s + b$

$= \left[\frac{1}{2} a T^2 + V_b T\right]$

- T ka finite formula

$2s = \frac{1}{2} a T^2 \Rightarrow \frac{4s}{a} = T^2 \Rightarrow \boxed{T = \sqrt{\frac{4s}{a}}}$

- alternate formula for T if s_1 & s_2 are different

$T = \sqrt{\frac{2(s_1 + s_2)}{a}}$ $\quad d_2 = s_1 + s_2 + b$

- V_b ki overtaken speed ka formula

• Jab V_b nai diya h speed of overtaken vehicle tab $V_b = (V_a - 4.5)\ ms^{-1}$

$V_b = (V_a - 16)$ kmph.

OSD
- formula for two way traffic in terms of d.
- for one way traffic

Note: for two way traffic

$OSD = d_1 + d_2 + d_3$

for one way traffic

$OSD = d_1 + d_2$

Overtaking zone length.
- Min length
- Desirable length

Overtaking zone length.

. Minimum length of overtaking zone: $3 \times OSD$. (matlab jo formulae se OSD aya usta 3 times)

. Desirable length of overtaking zone $= 5 \times OSD$

→ Minimum OSD ko aur kya bolte hai?

. Minimum OSD is also called safe passing distance.

→ agar minimum overtaking length possible nahi toh kitna dege

. if minimum or desirable overtaking zone length is not possible then provide $2 \times SSD$ (overtaking zone length)